

Project Development Phase

Heart Disease Analysis (Heart_new2 Dataset)

Date	11 July 2026
Team ID	
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Data Preprocessing Steps

- Step 1: Import the Heart_new2 dataset into Tableau and verify all fields (HeartDisease, BMI, Smoking, AlcoholDrinking, Stroke, PhysicalHealth, MentalHealth, DiffWalking, Sex, AgeCategory, Race, Diabetic, PhysicalActivity, GenHealth, SleepTime, Asthma, KidneyDisease, SkinCancer).
- Step 2: Check for missing values and handle null or blank records.
- Step 3: Remove duplicate patient records to ensure accurate analysis.
- Step 4: Validate data types for numeric and categorical fields.
- Step 5: Verify data consistency and standardize categorical values.
- Step 6: Create calculated fields such as Total Patients, Heart Disease Cases, Heart Disease Rate (%), Average BMI and Average Sleep Time.
- Step 7: Finalize the cleaned dataset for Tableau Dashboard and Story.

Business Questions with Visualization

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1	How many patient records are available after preprocessing?	Total Patients (KPI Card)
2	How many patients have heart disease?	Heart Disease Cases (KPI Card)
3	Which age group has the highest heart disease prevalence?	Age Category Analysis
4	How does BMI vary among heart disease patients?	BMI Analysis
5	How does smoking affect heart disease?	Smoking Analysis
6	How does diabetes relate to heart disease?	Diabetes Analysis
7	How does physical activity influence heart disease?	Physical Activity Analysis
8	How does general health differ between patients?	General Health Analysis

Dashboard KPIs

- Total Patients
- Heart Disease Cases
- Heart Disease Rate (%)
- Average BMI
- Average Sleep Time

Dashboard Visualizations

- Heart Disease Distribution
- Age Category Analysis
- Gender Analysis
- BMI Distribution
- Smoking vs Heart Disease
- Diabetes Analysis
- Physical Activity Analysis
- General Health Analysis

Note: Tableau screenshots from the original template were replaced with textual placeholders because actual dashboard images were not provided.